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Reg. No. | | | | | | | | |

Time : 3.00 hrs.

Quarterly Examination - 2026

BIOLOGY

Max. Marks : 70

PART - I (BIO - BOTANY)

SECTION - 1

I. Choose the correct answer: 8×1=8

1. First cell of male gametophyte in Angiosperm is
   a. microspore b. megaspore c. Nucleus d. Primary endosperm nucleus
2. Coleorhiza is found in a. paddy b. bean c. pea d. Tridax
3. Test cross involves
   a. crossing between two genotypes with recessive trait b. crossing between two F1 hybrids
   c. Crossing the F1 hybrid with a double recessive genotype
   d. crossing between two genotypes with dominant trait
4. Fruit colour in Squash is an example of
   a. recessive epistasis b. dominant epistasis c. complementary genes d. inhibitory genes
5. **Assertion (A):** Gamma rays are generally used to induce mutation in wheat varieties.
   Reason (R): Because they carry lower energy to non-ionize electrons from atoms.
   a. A and R are correct b. A is correct, R is wrong c. A is wrong, R is correct d. A and R are wrong
6. Plasmids are
   a. circular protein molecules b. required by bacteria c. tiny bacteria d. confer resistance to antibiotics
7. In which techniques Ethidium Bromide is used?
   a. Southern Blotting techniques b. Western Blotting techniques
   c. Polymerase chain reaction d. Agarose Gel electrophoresis
8. Solidifying agent used in plant tissue culture is
   a. nicotinic acid b. cobaltous chloride c. EDTA d. Agar

SECTION - 2

II. Answer any four of the following questions: 4×2=8

9. "Endothecium is associated with dehiscence of anther" - Justify the statement.
10. Draw and label the parts of the structure of ovule.
11. Give the names of the scientists who rediscovered Mendelism.
12. Define Nullisomy.
13. What is Bio-remediation?
14. Define Cybrid.

SECTION - 3

III. Answer any three of the following questions. (Q. No. 19 is compulsory): 3×3=9

15. Write short note on Anemophilous flowers.
16. What are the reasons for Mendel's success in his breeding experiment?
17. Write the uses of genetic mapping.
18. What do you know about the word pBR 322?
19. What are the advantages of Artificial seeds?

SECTION - 4

IV. Answer all the questions: 2×5=10

20. Explain the Polygonum type of embryosac development. (OR)
    Explain Incomplete dominance with an example.
21. Write the advantages and disadvantages of Bt Cotton. (OR)
    Write the application of plant tissue culture.

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BIO - ZOOLOGY

PART – A

I. **Choose the correct answer:** 8×1=8

1. Given example of Isogamy --------
   (a) Protozoa (b) Monocystis (c) Trichonympha (d) Actinophrys
2. Find the wrongly matched pair:
   (a) Bleeding Phase - fall in Oestrogen and Progesterone
   (b) Follicular phase - rise in Oestrogen
   (c) Luteal phase - rise in FSH level
   (d) Ovulatory phase - LH Surge
3. Find the correctly matched pair:
   (a) Gonorrhoea - Candida albicans (b) Genital herpes - Herpes Simplex Virus
   (c) Candidiasis - Trichomonas vaginalis (d) Trichomoniasis - Neisseria gonorrhoeae
4. What can be the blood group of offspring when both parents have AB blood group?
   (a) AB only (b) AB, B, A and O (c) A and O (d) A, B, and AB
5. An operon is --------
   (a) Cluster of structural genes with related function (b) Protein that suppresses gene expression
   (c) Protein that accelerates gene expression (d) Genes that switched other genes on (or) off
6. Modern man belongs to which period?
   (a) Cambrian (b) Cretaceous (c) Quaternary (d) Silurian
7. *Exo-erythrocytic* Schizogony of Plasmodium takes place in --------
   (a) RBC (b) WBC (c) Stomach (d) Liver
8. CO_2 is not released during:
   (a) Aerobic respiration in animals (b) Aerobic respiration in plants
   (c) Lactate fermentation (d) Alcoholic fermentation

PART - B

II. **Answer any Four Questions only:** 4×2=8

9. List out chemical alarm signals produced during inflammation.
10. Arrange the descent in human evolution.
11. Give reasons: Genetic code is Universal.
12. How are STDs transmitted?
13. What is the composition of semen?
14. What is Apolysis?

PART - C

III. **Answer any three of the following questions (Q. No: 19 is compulsory):** 3×3=9

15. Differentiate between regeneration in lizard and planaria
16. Write a short note on Lactational amenorrhea.
17. Differentiate between Divergent evolution and Convergent evolution with one example for each.
18. List the common withdrawal symptoms of drugs and alcohol abuse.
19. What is referred to as Industrial alcohol? Briefly describe its preparation.

PART - D

IV. **Answer the following questions:** 2×5=10

20. (a) Explain the different kinds of Syngamy in living organisms. **(OR)**
    (b) Explain the genetic basis of ABO blood grouping in man.
21. (a) Write the salient features of the Human Genome Project. **(OR)**
    (b) Explain the structure of Immunoglobulin with a suitable diagram.

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